Geography of Innovation in HealthTech

University-Industry Spillovers and Surgical Robotics Clusters

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Abstract

Digital information circulates globally, yet innovation in complex medical hardware remains highly localized. This report examines that paradox in surgical robotics and related HealthTech activities. Using a Scopus-based bibliometric corpus of publications from 2020 to 2024, it first maps the global geography of research production, then studies university–industry collaboration as an observable channel of knowledge spillovers, and finally analyzes the Strasbourg Nextmed ecosystem as a mechanism-revealing case study. The results show that scientific output is strongly concentrated in a limited number of countries and regional environments, that university–industry collaboration exhibits a pronounced domestic bias, and that international collaboration follows selective corridors rather than a frictionless global pattern. The Strasbourg case helps explain these findings by showing how proximity reduces coordination costs, supports tacit knowledge transfer through training and clinical practice, and shortens the path from research to commercialization. Taken together, the evidence suggests that in HealthTech, geography still matters because innovation depends not only on codified information, but also on repeated interaction among clinicians, engineers, firms, and translational institutions.

1 Introduction

A central promise of the digital economy is that knowledge can travel quickly across space. In principle, this should reduce the importance of geography for innovation. Yet in HealthTech, and especially in surgical robotics, the opposite pattern remains visible: research, training, clinical validation, and entrepreneurial activity cluster in a limited number of places rather than spreading evenly across regions.

This report asks a simple but important question: does complex HealthTech knowledge travel easily at a distance, or does it remain geographically sticky because key forms of expertise are tacit, interdisciplinary, and embedded in clinical practice? The issue matters for the economics of innovation because surgical robotics is not a purely digital industry. It combines software, hardware, imaging, regulation, clinical testing, and user training. As a result, the relevant knowledge is produced and used by heterogeneous actors who must solve problems together.

The report argues that geographic proximity remains economically important in this field. The intuition is consistent with research on localized knowledge spillovers, tacit knowledge, and the Triple Helix relationship between universities, industry, and public institutions (Audretsch and Feldman 1996; Gertler 2003; Storper and Venables 2004; Etzkowitz and Leydesdorff 2000). When knowledge is difficult to codify and validation depends on repeated interaction, co-location can reduce coordination costs and accelerate learning.

To examine this argument, the report proceeds in three steps. Chapter 1 maps the macro geography of surgical-robotics-related research using bibliometric evidence. Chapter 2 studies university–industry collaboration patterns to test whether proximity shapes observable knowledge exchange. Chapter 3 then turns to Strasbourg and the Nextmed ecosystem to show how these mechanisms operate on the ground. The report therefore moves from global structure to collaboration patterns and finally to a case that helps explain why clustering persists in a highly connected economy.

2 Chapter 1: Mapping the Clusters of Surgical Robotics

2.1 Data, method, and analytical framework

The first step is to establish whether innovation in surgical robotics and related HealthTech activity is actually spatially concentrated. To do so, we built a bibliometric dataset from **Scopus**, covering publications from 2020 to 2024.[1] This time window captures recent developments in a fast-moving field while preserving a sufficiently broad corpus for geographic comparison.

Bibliometric data are useful here for three reasons. First, they provide comparable evidence on where research is being produced. Second, affiliation metadata make it possible to connect publications to institutional actors and locations. Third, they offer a tractable basis for spatial analysis. At the same time, publication data do not capture all innovation. Some proprietary corporate development, early-stage prototyping, or informal clinical

[1] The Scopus query combines title, abstract, and keyword terms related to surgical robotics, robot-assisted surgery, teleoperation, medical robotics, image-guided intervention, and medical-device terminology, together with affiliation filters for academic institutions and corporate actors, and restricts the sample to publications from 2020 to 2024.

experimentation remains invisible in the academic record. For that reason, the bibliometric results are interpreted as evidence on the visible research geography of the field rather than as a complete measure of all innovation activity.

The analysis draws on work showing that knowledge spillovers are often localized rather than spatially neutral (Jaffe, Trajtenberg, and Henderson 1993). This matters especially in HealthTech, where tacit routines, clinical judgment, engineering constraints, and regulatory learning are difficult to transmit through documents alone (Gertler 2003; Storper and Venables 2004). The Triple Helix perspective further suggests that innovation emerges from dense interactions among universities, firms, and public institutions (Etzkowitz and Leydesdorff 2000). If that is true, surgical robotics should not be geographically diffuse. It should cluster where these capabilities are jointly present.

2.2 Global concentration of publications

The global map of publications confirms that scientific output in the corpus is highly uneven rather than broadly distributed. The United States leads by a clear margin, followed by China. Germany and the United Kingdom stand out within Europe, while Japan, France, Canada, and Australia also appear as important contributors. By contrast, large parts of Latin America, Africa, and Asia remain weakly represented.

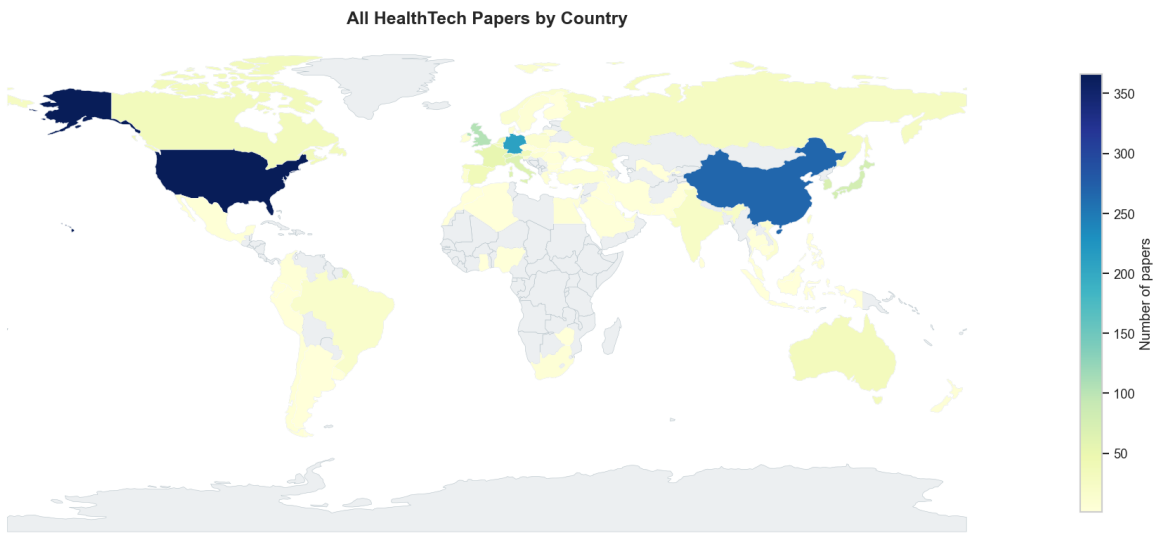

Figure 1: Global distribution of publications in the surgical-robotics-related HealthTech corpus by country.

This pattern already points toward concentration rather than dispersion. Output does not simply follow the existence of global digital communication. It is strongest where advanced research capacity, hospitals, engineering capabilities, and translational infrastructures are already present. In that sense, the publication map supports the idea that complex medical innovation is embedded in institutional environments rather than

detached from place.

2.3 Spatial clustering beyond visual inspection

To move beyond visual interpretation, the report evaluates spatial clustering with Moran's I and local indicators of spatial association (LISA) (Anselin 1995). The results show statistically significant positive spatial autocorrelation.

Table 1: Global Moran's I for the spatial distribution of publications in the corpus

Statistic	Value
Moran's I	0.3446
Expected I	-0.0057
Z-score	7.2782
P-value	0.0010

The positive and significant Moran's I indicates that publication output is not randomly distributed.[2] High-output countries tend to be located near other strong performers, while lower-output countries are often grouped in weaker-performing regions.

The local cluster structure refines this result. The LISA map identifies high–high clusters, especially in Western Europe and South Korea, suggesting consolidated regional cores of innovation. It also reveals several high–low outliers, such as China and South Africa, which behave more like innovation islands than as parts of broader regional systems. The United States appears as non-significant under this spatial specification, which does not indicate weak output, but rather that its performance is not statistically clustered with neighboring countries in the same way.

Taken together, Figure 1, Table 1, and Figure 2 show that the global geography of research is structured and unequal. Even in a digitally connected world, innovation in surgical robotics remains concentrated in a limited set of places with the institutional depth to sustain it. This macro evidence sets up the next question: if innovation is geographically concentrated, is collaboration also shaped by proximity?

3 Chapter 2: Proximity and Knowledge Spillovers

3.1 University–industry co-authorship as an observable mechanism

A useful way to study knowledge spillovers is to observe collaborations between universities and firms. Co-authorship does not capture every form of interaction, but it provides a clear and transparent indicator of coordinated research across organizational boundaries.

[2]This should be interpreted as evidence of country-level clustering, not as definitive proof of a causal spillover mechanism by itself.

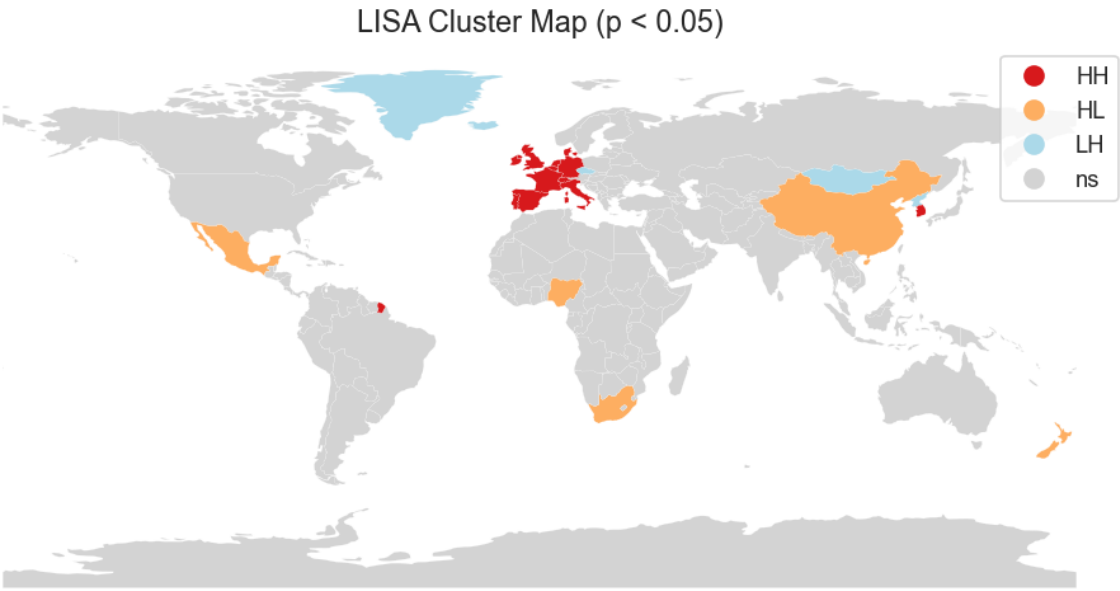

Figure 2: LISA cluster map for the spatial distribution of publications in the corpus.

In sectors characterized by uncertainty and interdisciplinary problem-solving, such collaboration is unlikely to be random. If proximity reduces communication costs and facilitates repeated interaction, one should expect university–industry collaboration to display a geographic structure as well (Boschma 2005; Gertler 2003; Storper and Venables 2004).

The dataset contains 662 valid university–industry co-authored publications. A first result is the presence of a strong domestic bias. In 84.4% of cases, the academic and corporate partners are located in the same country, while only 15.6% involve cross-country collaboration.[3] This is a substantial imbalance. It suggests that even in a global field, collaborative research remains strongly anchored in national environments.

The country-level distribution of these publications is also uneven. The United States and China dominate in absolute volume, each with collaboration shares around three quarters of their output. At the same time, countries with smaller overall publication systems, including Japan, Canada, and Qatar, show particularly high shares of university–industry collaboration. This suggests that institutional specialization can generate strong local integration even outside the largest systems.

Figure 3 therefore supports a first interpretation: collaboration is neither spatially neutral nor evenly globalized. It is concentrated in a limited number of countries and remains strongly tied to domestic research ecosystems.

[3]Here, *home bias* refers to the tendency of firms to collaborate with universities located in the same country rather than abroad. Country-level co-location is used as a practical proxy for proximity, even though it does not capture regional distance within countries.

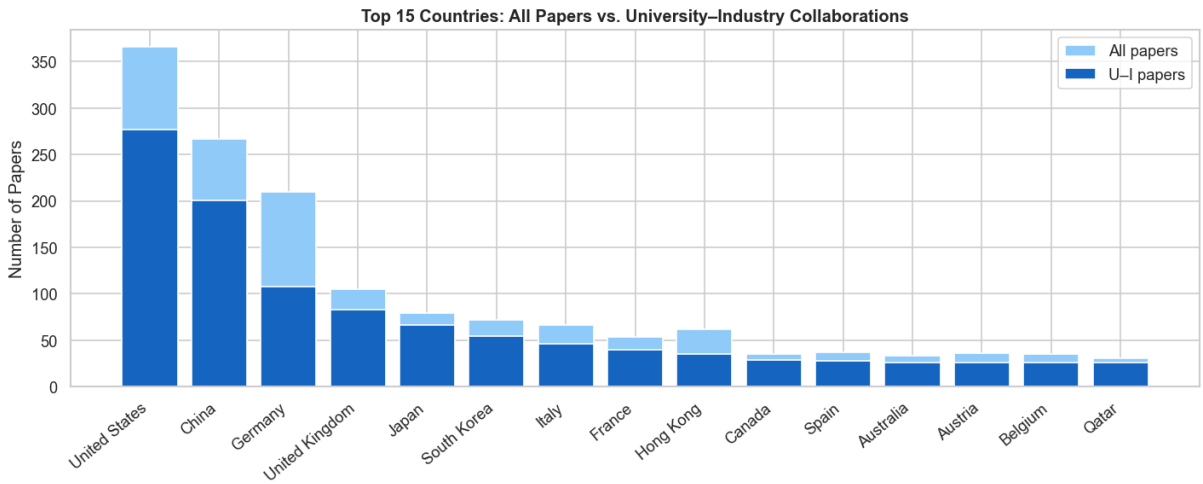

Figure 3: University–industry co-authored publications by country among the top 15 countries in the sample.

3.2 Selective international corridors

Domestic bias does not mean that international collaboration is absent. It means that it is selective. Among 245 cross-country university–industry pairs, a small number of bilateral corridors account for a disproportionate share of observed international ties. The strongest corridor links Italian universities with Spanish companies, with 98 collaborations. Spain also appears repeatedly as an industrial partner for universities in South Korea, China, the United States, and the United Kingdom. Other important corridors include Italy–United Kingdom, Italy–Germany, China–Germany, and United States–Germany.

Figure 4 shows that cross-border knowledge exchange is structured rather than frictionless. International collaboration does not form a smooth global network. Instead, it clusters around repeated relationships that likely reflect prior institutional ties, complementary capabilities, regulatory familiarity, and accumulated trust. Put differently, distance can be overcome, but usually through stable channels rather than open-ended global reach.

3.3 Interpreting the collaboration evidence

The combined evidence points toward a consistent conclusion. First, collaboration is heavily domestic. Second, when it becomes international, it follows a narrow set of strong corridors rather than a diffuse pattern. These findings fit the broader innovation-literature argument that proximity matters when knowledge is tacit, interdisciplinary, and costly to coordinate (Audretsch and Feldman 1996; Boschma 2005; Gertler 2003). In surgical robotics, collaboration often requires feedback between clinicians, engineers, firms, and research teams. Such interaction is easier to sustain when organizational and geographic distance are reduced.

This does not imply that all valuable knowledge must be local. The field clearly

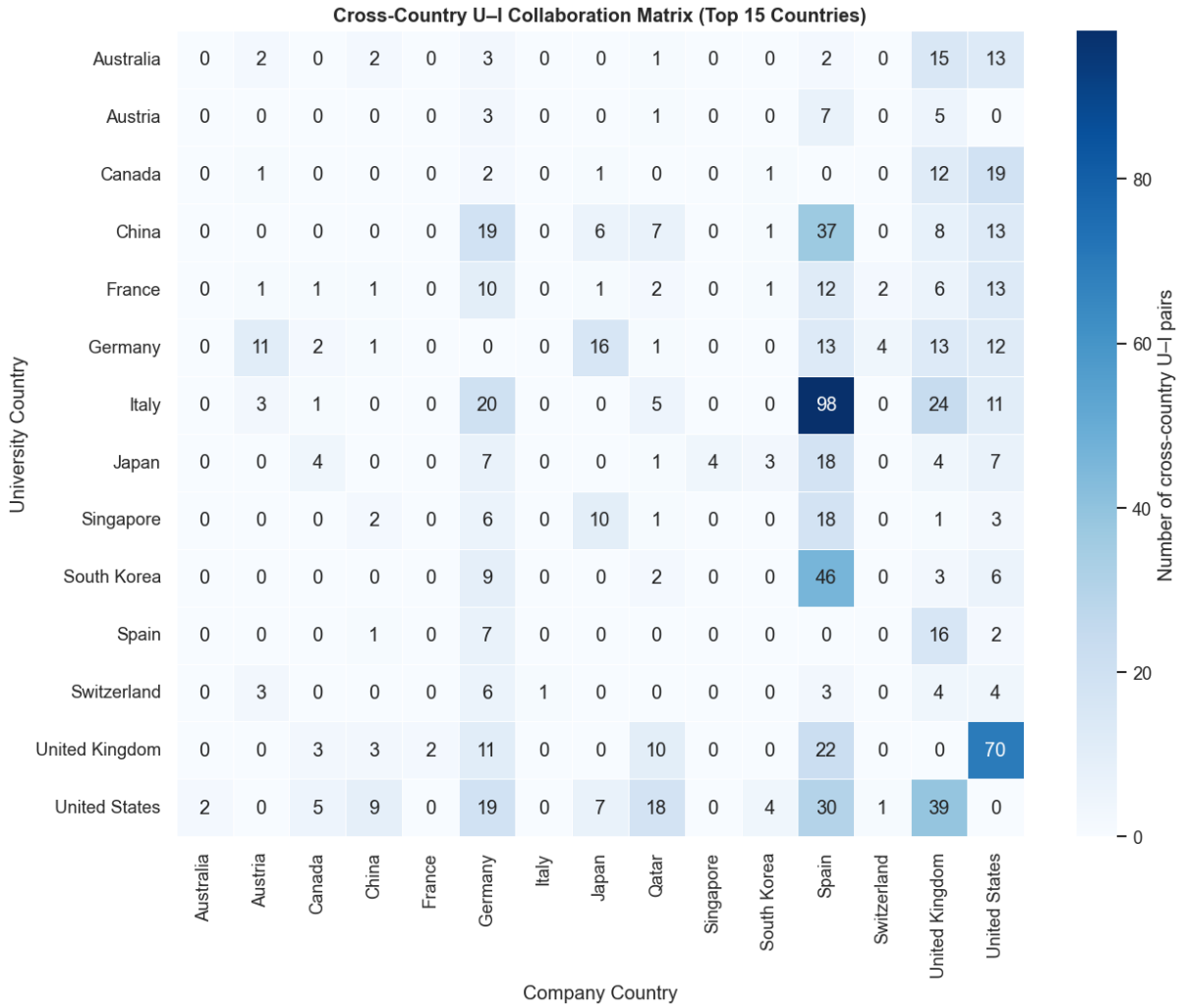

Figure 4: Cross-country university–industry collaboration matrix showing the main international collaboration corridors.

contains international exchange. However, the evidence suggests that global connectivity does not eliminate the value of co-location. Instead, proximity appears to shape the default structure of collaboration, while international links emerge as selective extensions of already organized systems. The macro concentration observed in Chapter 1 is therefore matched by a meso-level pattern in which observable knowledge exchange also remains geographically structured.

These results motivate a closer look at how proximity actually operates in practice. Strasbourg is especially useful for that purpose because it allows the report to move from statistical regularities to concrete institutional mechanisms.

4 Chapter 3: The Strasbourg Nextmed Ecosystem as an Explanatory Case Study

4.1 Why Strasbourg is analytically useful

Strasbourg is not simply another productive city in the publication map. It is a compact and highly coordinated HealthTech ecosystem in which hospitals, research laboratories, translational institutes, training centers, incubators, and firms are physically close to one another. That makes it a strong case for studying how localized spillovers may work.

The case is particularly relevant for three reasons. First, Strasbourg is specialized in image-guided intervention, minimally invasive therapies, and surgical innovation rather than relying only on metropolitan scale. Second, the cluster combines the key stages of the innovation chain within a limited urban perimeter. Third, the local ecosystem is documented by both institutional sources and visible research activity, which makes it possible to connect organizational structure to the mechanisms suggested by the macro analysis.

4.2 Institutional density and the local innovation chain

Nextmed is organized around the Strasbourg Civil Hospital site and presents itself as a place-based health-innovation campus (Nextmed Strasbourg 2026b; Nextmed Strasbourg 2026d; Ville et Eurométropole de Strasbourg 2026a). Its analytical importance lies in institutional density. Clinical care, engineering research, translational medicine, training, technology transfer, and entrepreneurial support are not dispersed across distant locations. They are compressed into a shared environment.

Several institutions anchor this structure. **IRCAD** is an internationally recognized training and experimentation center in minimally invasive and robotic surgery, reportedly training more than 8,800 surgeons per year with the participation of more than 800 experts (IRCAD 2026). **IHU Strasbourg** explicitly links patient care, translational research, education, and technology transfer (IHU Strasbourg 2026; Nextmed Strasbourg 2026c).

Figure 5: Annotated map of the Nextmed campus. The figure highlights the spatial concentration of the cluster on the Strasbourg Civil Hospital site and shows the immediate proximity between clinical, research, and entrepreneurial infrastructures.

ICube provides the engineering base in imaging, robotics, AI, and control (Université de Strasbourg 2026; Nextmed Strasbourg 2026c). On the clinical side, the **Hôpitaux Universitaires de Strasbourg** keep innovation tied to real therapeutic practice (Hôpitaux Universitaires de Strasbourg 2026b; Hôpitaux Universitaires de Strasbourg 2026a). Around these actors, entrepreneurial spaces such as **pH8** and **eXplora**, together with transfer and incubation organizations such as **Conectus**, **SEMIA**, and **Quest for Health**, reduce the distance between research and firm creation (Ville et Eurométropole de Strasbourg 2026b; Nextmed Strasbourg 2023; Nextmed Strasbourg 2026a; Conectus 2026a; Conectus 2026b).

What emerges from Figure 5 and Table 2 is not just a list of institutions, but a locally integrated innovation chain. This gives Strasbourg its explanatory value.

4.3 Mechanism 1: reducing coordination costs

The first mechanism is the reduction of coordination costs. In HealthTech, relevant knowledge is distributed across professions that do not share the same vocabulary, rhythm, or criteria of validation. Surgeons identify procedural bottlenecks, engineers translate them into technical constraints, imaging specialists refine data requirements, and firms evaluate product viability. When these actors are geographically separated, interaction becomes slower, more formal, and more fragmented. Co-location changes that.

Figure 6 makes this logic visible. The economic value of the cluster lies less in the mere existence of multiple actors than in the density of the links among them. Official materials

Table 2: Institutional roles within the Strasbourg Nextmed ecosystem and the economic relevance of proximity

Institution	**Primary role in the innovation chain**	**Why physical proximity matters**
IRCAD	International training center and experimentation hub for minimally invasive and robotic surgery.	Concentrates embodied surgical know-how and enables frequent feedback between expert users and innovators.
IHU Strasbourg	Translational institute linking care, research, education, and technology transfer.	Shortens the path between clinical problem identification, prototyping, validation, and transfer.
ICube	Engineering research base in robotics, imaging, AI, and medical technologies.	Supplies nearby technical capabilities required to convert clinical needs into devices and algorithms.
HUS / Civil Hospital	Clinical environment where needs, constraints, and validation conditions emerge.	Anchors innovation in real therapeutic practice and supports recurrent testing loops.
Nextmed	Campus-level coordinator organizing co-location and visibility.	Makes repeated formal and informal interaction cheaper and more likely.
pH8 / eXplora	Nursery and firm-hosting infrastructure.	Keeps startups embedded in the research and clinical environment during early development.
Conectus / SEMIA / Quest for Health	Technology-transfer and venture-support actors.	Reduces institutional friction between public research and commercialization.

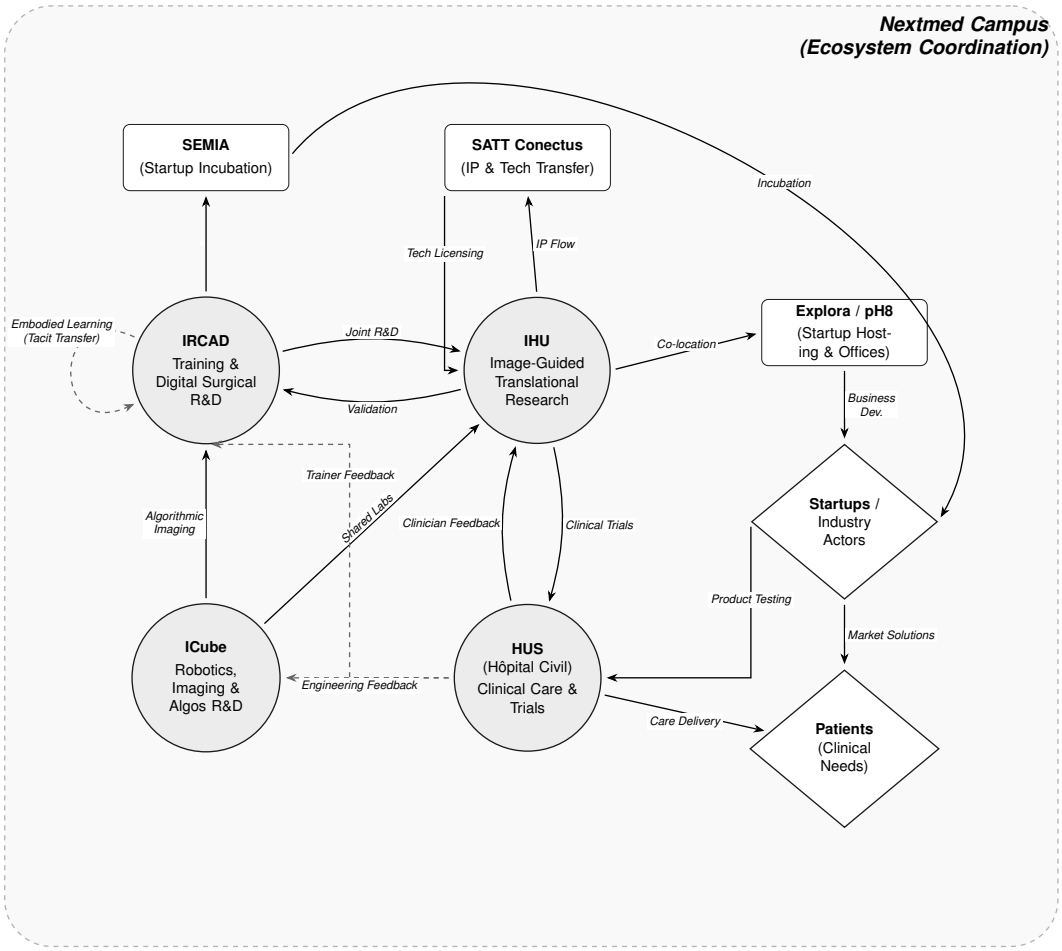

Figure 6: Institutional structure of the Nextmed Strasbourg ecosystem. The diagram maps the principal organizational actors of the cluster and the major flows linking them, including joint R&D, clinical validation, training, incubation, and technology transfer.

repeatedly describe Nextmed as a space designed to connect medical, scientific, and entrepreneurial competences directly (Nextmed Strasbourg 2026b; Nextmed Strasbourg 2026d; Nextmed Strasbourg 2023). This is precisely the kind of setting in which face-to-face problem-solving retains value (Boschma 2005; Storper and Venables 2004).

4.4 Mechanism 2: tacit knowledge transfer through training and clinical practice

The second mechanism is the circulation of tacit knowledge. In this field, some of the most valuable knowledge is not fully codified. Surgical judgment, ergonomic feedback, instrumentation constraints, and the practical integration of imaging with action are learned through observation, demonstration, correction, and repetition.

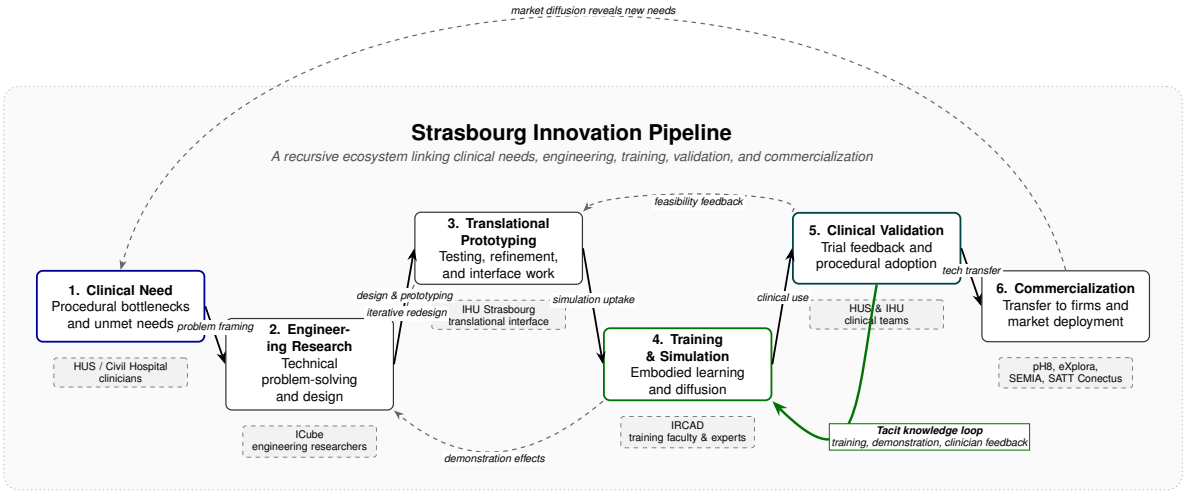

Figure 7: Innovation pipeline within the Strasbourg HealthTech ecosystem. The figure depicts a recursive sequence linking clinical need identification, engineering research, translational prototyping, training and simulation, clinical validation, and commercialization.

IRCAD is central to this mechanism because it functions not only as a research institution, but also as a site of embodied learning (IRCAD 2026). IHU Strasbourg reinforces this dynamic by linking education, clinical practice, and translational research within the same organizational setting (IHU Strasbourg 2026). Figure 7 captures the resulting process: innovation does not move linearly from laboratory to market, but recursively through feedback loops between need identification, design, simulation, testing, and refinement.

This helps explain why proximity matters more here than in highly codified sectors. In surgical robotics, innovation depends on repeated interaction with expert users under clinical constraints. Under those conditions, proximity is not merely convenient. It becomes part of the production process itself.

4.5 Mechanism 3: accelerating translation and commercialization

The third mechanism is the shortening of the path from research to market. A persistent difficulty in university–industry relations is that promising results remain trapped in laboratories while firms struggle to access credible testing environments. Strasbourg addresses that problem by organizing research, validation, and venture support within the same ecosystem.

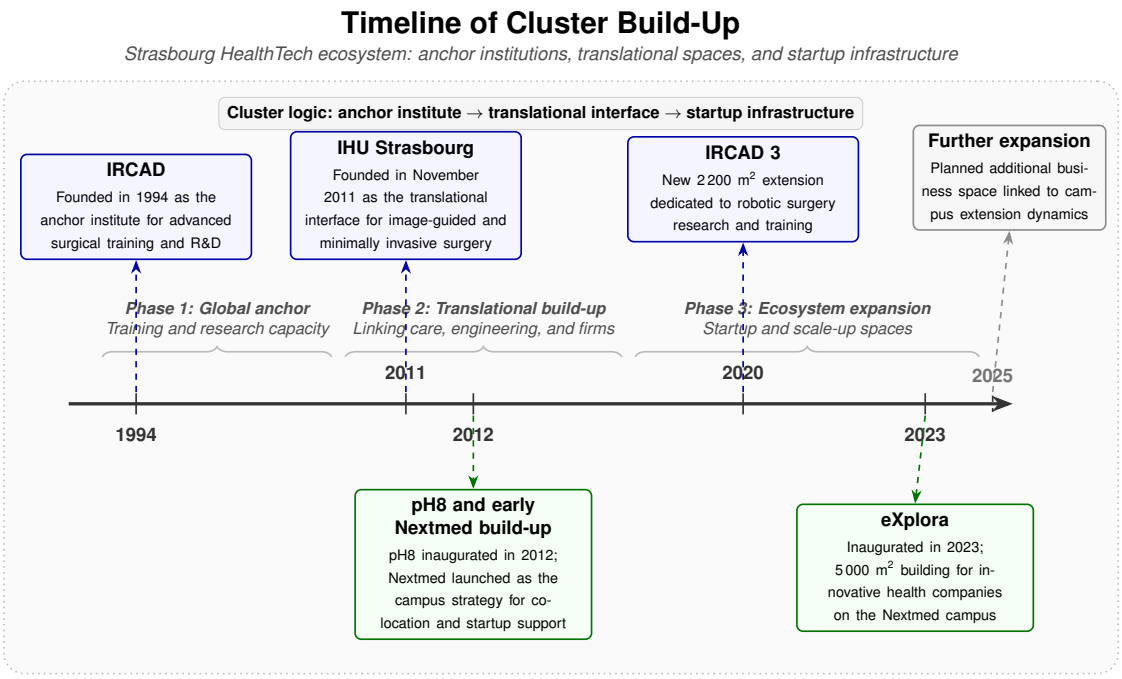

Figure 8: Timeline of cluster build-up in the Strasbourg HealthTech ecosystem. The chronology emphasizes the cumulative institutional layering through which the cluster connected surgical training, translational research, engineering capacity, and firm-hosting infrastructure.

Figure 8 shows that this capacity emerged cumulatively. Strasbourg did not become a cluster overnight. It was built through institutional layering that progressively linked training, engineering, translational medicine, and entrepreneurial support. Spaces such as pH8 and eXplora are important precisely because they allow young firms to remain physically close to hospitals, labs, and expert users (Ville et Eurométropole de Strasbourg 2026b; Nextmed Strasbourg 2023). Conectus, SEMIA, and related organizations reduce the legal and organizational frictions that often slow technology transfer (Nextmed Strasbourg 2026a; Conectus 2026b).

A concrete illustration is the CAMMA team, a joint IHU Strasbourg–ICube initiative working on AI-assisted surgery (IHU Strasbourg 2021b; IHU Strasbourg 2023). Its work on surgical video analysis and the "critical view of safety" emerged from real clinical problems in laparoscopic practice and was developed through close interaction among

clinicians, engineers, and hospital-based environments (Mascagni et al. 2020; Mascagni et al. 2024; IHU Strasbourg 2021a). The later creation of **Scialytics** as a spin-off shows how a local ecosystem can support movement from clinical need to research, validation, and entrepreneurial exploitation (IHU Strasbourg 2025). The point is not that Strasbourg is unique in producing spin-offs, but that its institutional configuration increases the probability that translation will occur.

4.6 What the case explains

As a case study, Strasbourg is not statistically representative of all HealthTech systems. It is best understood as a strategically selected example that makes mechanisms visible. Its value lies in showing how the patterns observed in Chapters 1 and 2 can arise in practice. The cluster reduces coordination costs, supports tacit learning, and compresses the sequence from invention to validation and commercialization. It therefore gives concrete meaning to the idea of localized university–industry spillovers.

5 Conclusion

This report began from a puzzle of the digital economy: if information travels globally, why does innovation in surgical robotics remain so geographically concentrated? The answer developed here is that complex HealthTech innovation depends on more than codified knowledge. It requires repeated interaction among clinicians, engineers, firms, and translational institutions, as well as access to training environments and clinical validation.

The macro analysis showed that research output is strongly concentrated and spatially clustered. The collaboration analysis showed that university–industry ties are heavily domestic and that cross-border collaboration follows selective corridors rather than a frictionless global pattern. The Strasbourg case then explained why these patterns persist by revealing three mechanisms through which proximity matters: lower coordination costs, richer tacit knowledge transfer, and faster translation from research to commercialization.

The broader implication is that the geography of innovation has not disappeared in the digital era. In sectors such as surgical robotics, digital connectivity complements local ecosystems rather than replacing them. HealthTech innovation remains local not because information cannot travel, but because some of its most valuable forms still depend on shared infrastructures, repeated contact, and place-based institutional coordination.

References

Anselin, L. (1995). "Local Indicators of Spatial Association—LISA". In: *Geographical Analysis* 27.2, pp. 93–115.

Audretsch, D. B. and M. P. Feldman (1996). "R&D Spillovers and the Geography of Innovation and Production". In: *The American Economic Review* 86.3, pp. 630–640.

Boschma, R. (2005). "Proximity and Innovation: A Critical Assessment". In: *Regional Studies* 39.1, pp. 61–74.

Conectus (2026a). *SATT Conectus | Transfert de technologies et collaborations*. URL: https://www.conectus.fr/ (visited on 03/07/2026).

— (2026b). *When the Researcher's Invention Becomes a Startup*. URL: https://www.conectus.fr/en/when-researchers-invention-becomes-startup (visited on 03/07/2026).

Etzkowitz, H. and L. Leydesdorff (2000). "The Dynamics of Innovation: From National Systems and "Mode 2" to a Triple Helix of University–Industry–Government Relations". In: *Research Policy* 29.2, pp. 109–123.

Gertler, M. S. (2003). "Tacit Knowledge and the Economic Geography of Context, or The Undefinable Tacitness of Being (There)". In: *Journal of Economic Geography* 3.1, pp. 75–99.

Hôpitaux Universitaires de Strasbourg (2026a). *Hôpital civil*. URL: https://www.chru-strasbourg.fr/hopital/hopital-civil/ (visited on 03/07/2026).

— (2026b). *Les Hôpitaux Universitaires de Strasbourg*. URL: https://www.chru-strasbourg.fr/ (visited on 03/07/2026).

IHU Strasbourg (2021a). *Artificial Intelligence for Safer Surgery*. URL: https://www.ihu-strasbourg.eu/artificial-intelligence-for-safer-surgery/?lang=en (visited on 03/07/2026).

— (2021b). *Research Scientist Recruitment Notice for CAMMA*. URL: https://www.ihu-strasbourg.eu/wp-content/uploads/IHU_recrutement_Research_Scientist_EN.pdf (visited on 03/07/2026).

— (2023). *Prof. Nicolas Padoy Awarded €1.95 Million ERC Grant for Ambitious Digital Surgery Project COMP-SURG*. URL: https://www.ihu-strasbourg.eu/prof-nicolas-padoy-awarded-e1-95-million-erc-grant-for-ambitious-digital-surgery-project-compsurg/?lang=en (visited on 03/07/2026).

— (2025). *Scialytics: IHU Spin-Off Is Building the First AI-Powered End-to-End Ecosystem to Measure and Reduce Surgical Risk*. URL: https://www.ihu-strasbourg.eu/scialytics-une-spin-off-de-ihu-strasbourg/?lang=en (visited on 03/07/2026).

— (2026). *IHU Strasbourg | Homepage*. URL: https://www.ihu-strasbourg.eu/?lang=en (visited on 03/07/2026).

IRCAD (2026). *IRCAD: MIS Training Center*. URL: https://www.ircad.fr/ (visited on 03/07/2026).

Jaffe, A. B., M. Trajtenberg, and R. Henderson (1993). "Geographic Localization of Knowledge Spillovers as Evidenced by Patent Citations". In: *The Quarterly Journal of Economics* 108.3, pp. 577–598.

Mascagni, P., D. Alapatt, A. Lapergola, A. Vardazaryan, J.-P. Mazellier, B. Dallemagne, D. Mutter, and N. Padoy (2024). "Early-Stage Clinical Evaluation of Real-Time Artificial Intelligence Assistance for Laparoscopic Cholecystectomy". In: *British Journal of Surgery* 111.1, znad353. DOI: 10.1093/bjs/znad353.

Mascagni, P., A. Vardazaryan, D. Alapatt, T. Urade, T. Emre, S. Perretta, D. Mutter, and N. Padoy (2020). "Formalizing Video Documentation of the Critical View of Safety in Laparoscopic Cholecystectomy: A Step Towards Artificial Intelligence Assistance to Improve Surgical Safety". In: *Surgical Endoscopy* 34.6, pp. 2709–2714. DOI: 10.1007/s00464-019-07054-5.

Nextmed Strasbourg (2023). *Inauguration du bâtiment eXplora*. URL: https://nextmed-strasbourg.eu/2023/07/27/inauguration-du-batiment-explora/ (visited on 03/07/2026).

— (2026a). *Accélérateurs de business*. URL: https://nextmed-strasbourg.eu/offre-de-services/accelerateurs-de-business/ (visited on 03/07/2026).

— (2026b). *Accueil*. URL: https://nextmed-strasbourg.eu/ (visited on 03/07/2026).

— (2026c). *Centres de R&D*. URL: https://nextmed-strasbourg.eu/innovateurs/centres-de-rd/ (visited on 03/07/2026).

— (2026d). *Nextmed, accélérateur d'innovation santé*. URL: https://nextmed-strasbourg.eu/decouvrez-nextmed/nextmed-innovation-sante/ (visited on 03/07/2026).

Storper, M. and A. J. Venables (2004). "Buzz: Face-to-Face Contact and the Urban Economy". In: *Journal of Economic Geography* 4.4, pp. 351–370.

Université de Strasbourg (2026). *Research laboratories | ICube*. URL: https://mathinfo.unistra.fr/en/research/research-laboratories/ (visited on 03/07/2026).

Ville et Eurométropole de Strasbourg (2026a). *Nextmed | Instrumental for Tomorrow's Healthcare*. URL: https://int.strasbourg.eu/nextmed (visited on 03/07/2026).

— (2026b). *pH8, la pépinière d'entreprises innovantes à Strasbourg*. URL: https://www.strasbourg.eu/ph8 (visited on 03/07/2026).